KISHORE KARANAM

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SUMMARY

Experienced Data Engineering professional with 3 years specializing in Big Data analytics, data visualization, business intelligence, and applied machine learning and AI. Proven ability to deliver impactful insights and scalable data solutions

WORK EXPERIENCE

Role:AWS Data Engineer — **Bloomington, IL**
Client: State Farm — **May 2024-DEC 2024**

- Collaborated with cross-functional teams to understand business requirements and design data solutions that improved data accessibility and operational efficiency.
- Built and managed scalable ETL pipelines using AWS Glue, Airflow, and PySpark, automating data ingestion and transformation from diverse sources into Amazon S3 and Redshift.
- Developed real-time and batch data workflows using Lambda, Kinesis, and Step Functions, enabling event-driven architectures and responsive analytics.
- Designed and maintained metadata using AWS Glue Data Catalog and Crawlers, optimizing query performance in Athena and reducing latency for business-critical reports.
- Created and tuned complex SQL queries in Athena and integrated them with Tableau dashboards to deliver actionable insights for stakeholders.
- Implemented data governance practices including encryption, partitioning, lifecycle policies, and archival strategies in Amazon S3 to ensure compliance and cost control.
- Supported CI/CD integration of data pipelines, provided on-call production support, and contributed to continuous improvements through performance tuning and documentation.

Technologies: AWS Glue, S3, Lambda, Kinesis, Athena, Airflow, PySpark, Databricks, Tableau, Step Functions, SQL, JSON, Parquet, Agile.

Role: Data Engineer — **Hyderabad, India**
Client: CA Technologies — **Aug 2021 – March 2023**

- Designed and implemented real-time data processing pipelines using AWS Lambda and Scala, optimizing SQL Server databases to handle up to 200% increased query throughput.
- Migrated on-premises SQL Server databases to AWS RDS with Multi-AZ deployments, improving performance by 40% while ensuring high availability and automated failover capabilities.
- Developed infrastructure as code (IaC) with AWS CloudFormation templates to automate deployment of scalable data architecture, implementing partitioning strategies and query optimization for enterprise-scale workloads.

Technologies: Python, SQL, AWS (S3, RDS, Redshift, EC2, Glue DataBrew), Spark, Apache Kafka, MySQL, MS SQL, Snowflake, Databricks, Tableau, Power BI, Git

Role: Data Analyst — **Hyderabad, India**
Client: Thomson Reuters — **Sep 2020 –June 2021**

- Performed comprehensive data analysis using SQL queries and Python to extract actionable insights from large datasets, creating interactive dashboards that increased departmental decision-making efficiency by 30%.
- Collaborated with cross-functional teams to identify business requirements and develop automated reporting solutions, translating complex data findings into clear visualizations for stakeholders.

Technologies: Python, SQL, Tableau, Power BI, Excel, Statistical Analysis, Data Visualization

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Databases and Big Data technologies:** MySQL, Spark, HDFS, Hive, Apache Kafka, MS SQL
- **Visualization & Version control tools:** Tableau, Power BI, Git
- **Cloud Services:** AWS(S3, RDS,Redshift, EC2, Glue DataBrew), Snowflake, Databricks,Google cloud Platform

PROJECTS

Agentic AI for Real-time Compliance Monitoring and NIST Cybersecurity Assessment

- Implemented an AI-powered NIST compliance monitoring system achieving 89-93% accuracy across varying dataset sizes.
- Developed an automated AWS Lambda workflow connecting SecureGPT analysis to JIRA ticket generation for real-time security response.
- Applied advanced prompt engineering techniques that successfully identified 10,000+ compliance violations in system logs without model retraining.

Brain Tumor Detection using Machine Learning

- By using machine learning, we built a binary classifier to detect brain tumors from MRI scan images.
- We built our classifier using transfer learning and obtained an accuracy of 96.5 percent and visualized our models overall performance.

Diamond Price Analysis

- Designed and developed an AI-powered model to predict diamond prices based on critical features such as carat, cut, clarity, and color, driving smarter pricing strategies.
- Built a scalable machine learning pipeline using **Python, Pandas, NumPy, and Scikit-learn**, achieving high prediction accuracy and reducing pricing discrepancies.
- Applied **exploratory data analysis** and **correlation mapping** to uncover hidden patterns and anomalies, leading to data-informed adjustments in pricing policies.
- Created interactive data visualizations with **Matplotlib**, **Seaborn**, and **Plotly**, enabling stakeholders to monitor trends and gain deeper market insights.
- Delivered a dynamic, insight-driven pricing framework that enhanced decision-making and contributed to improved profit margins.

EDUCATION

Master of Science in Data Analytics Engineering — **Fairfax, VA, USA**
George Mason University, GPA: 3.63 — Aug 2023 - May 2025

Bachelor of Technology in Computer Science and Engineering — **Visakhapatnam, AP, India**
GITAM Deemed University — July 2017 - July 2021

ARTICLES PUBLISHED

COVID-19 Cases Using Machine Learning, Link

Apple Inc.'s Financial Performance using Machine Learning, Link

A Data-Driven Analysis of Boston's Crime Rate, Link